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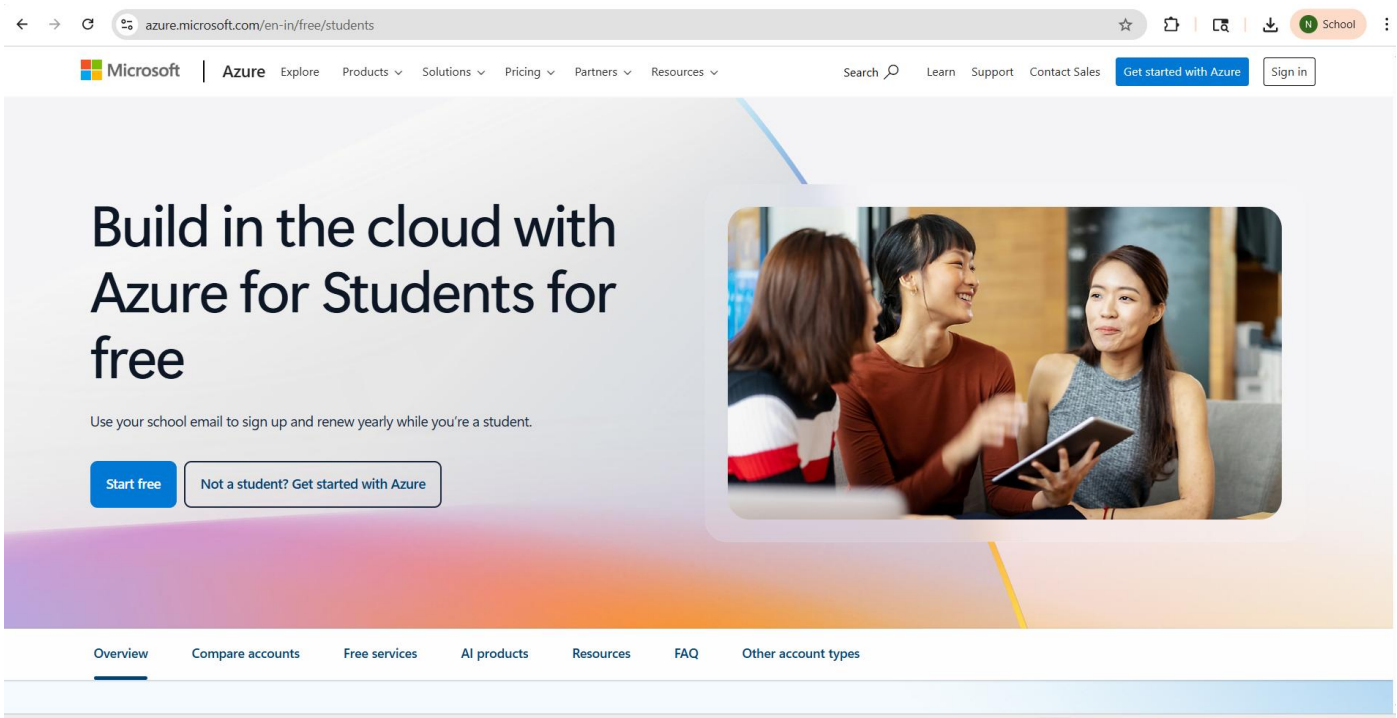
EXP NO: 1 DATE: 23/1/26	AZURE DEVOPS ENVIRONMENT SETUP

AIM:-

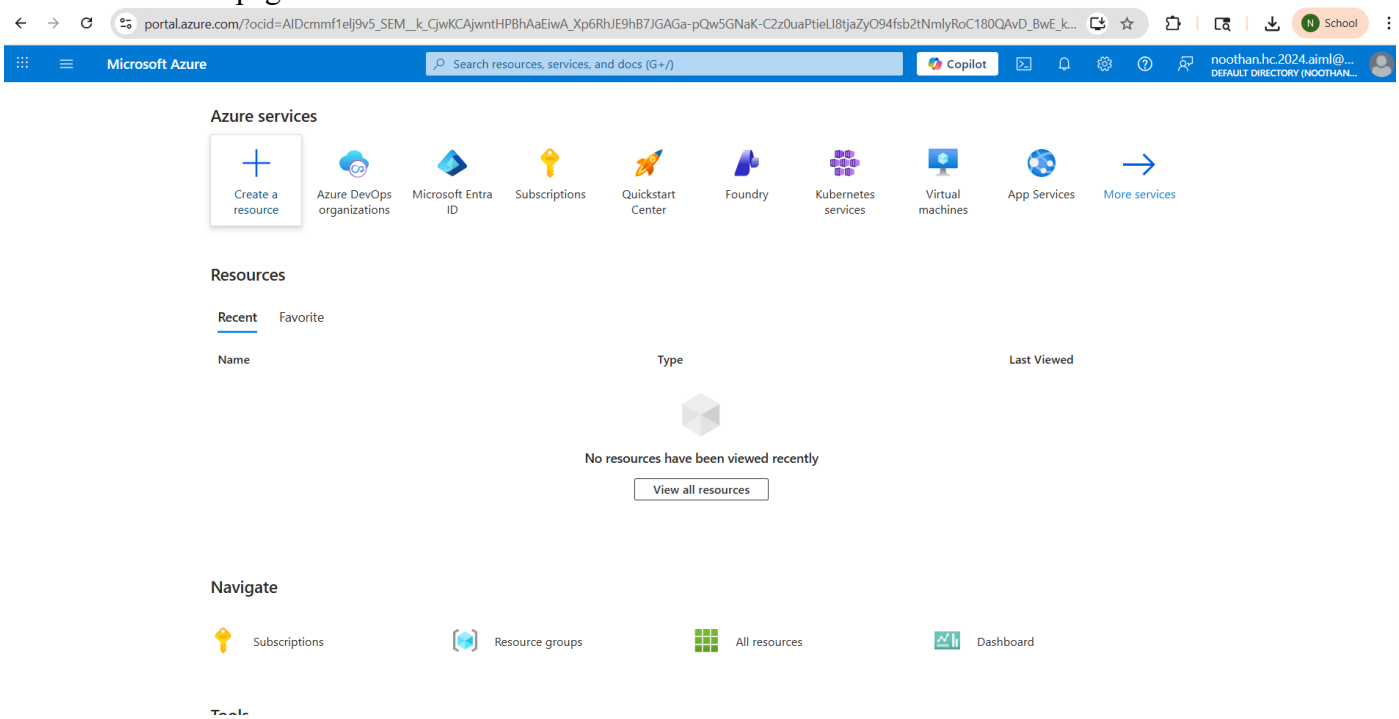
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION:-

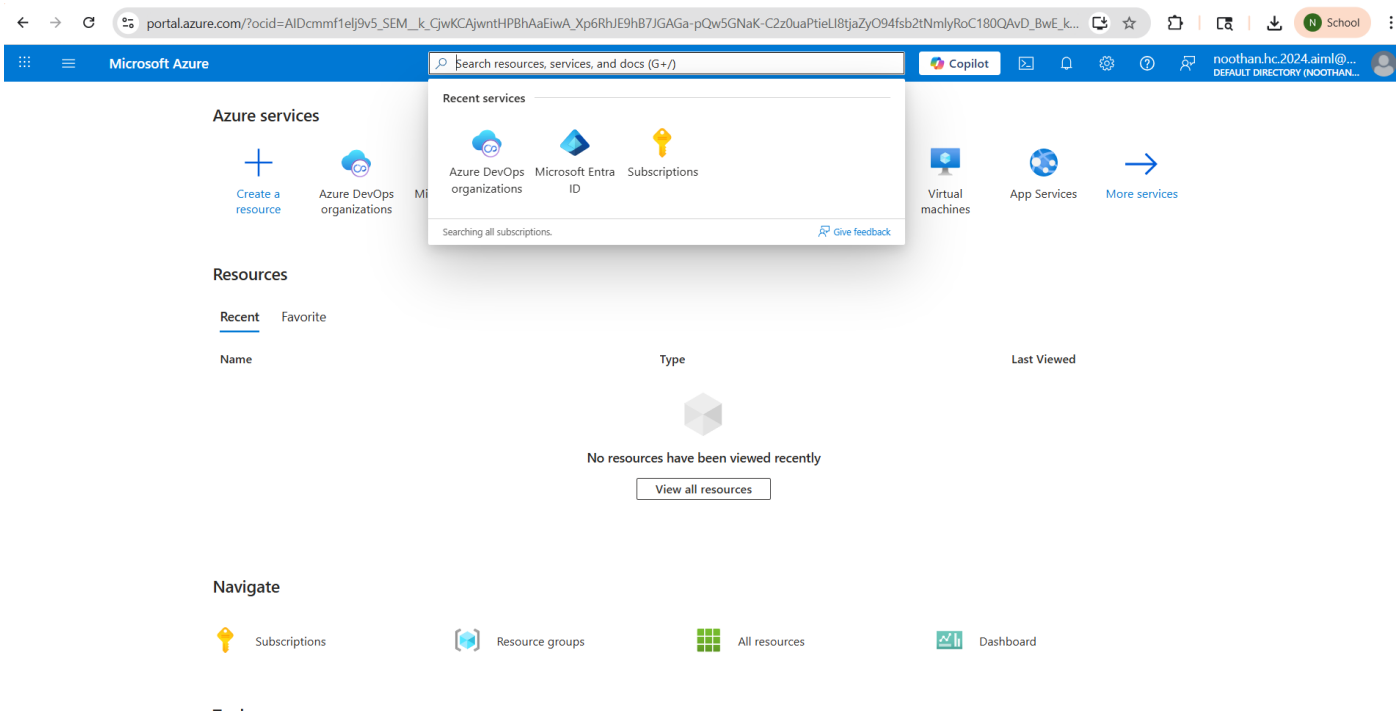
1. Open your web browser and go to the Azure website:
https://azure.microsoft.com/en-us/get- started/azure-portal.
Sign in using your Microsoft account credentials.
If you don't have a Microsoft account, you can create one here: https://signup.live.com/?lic=1



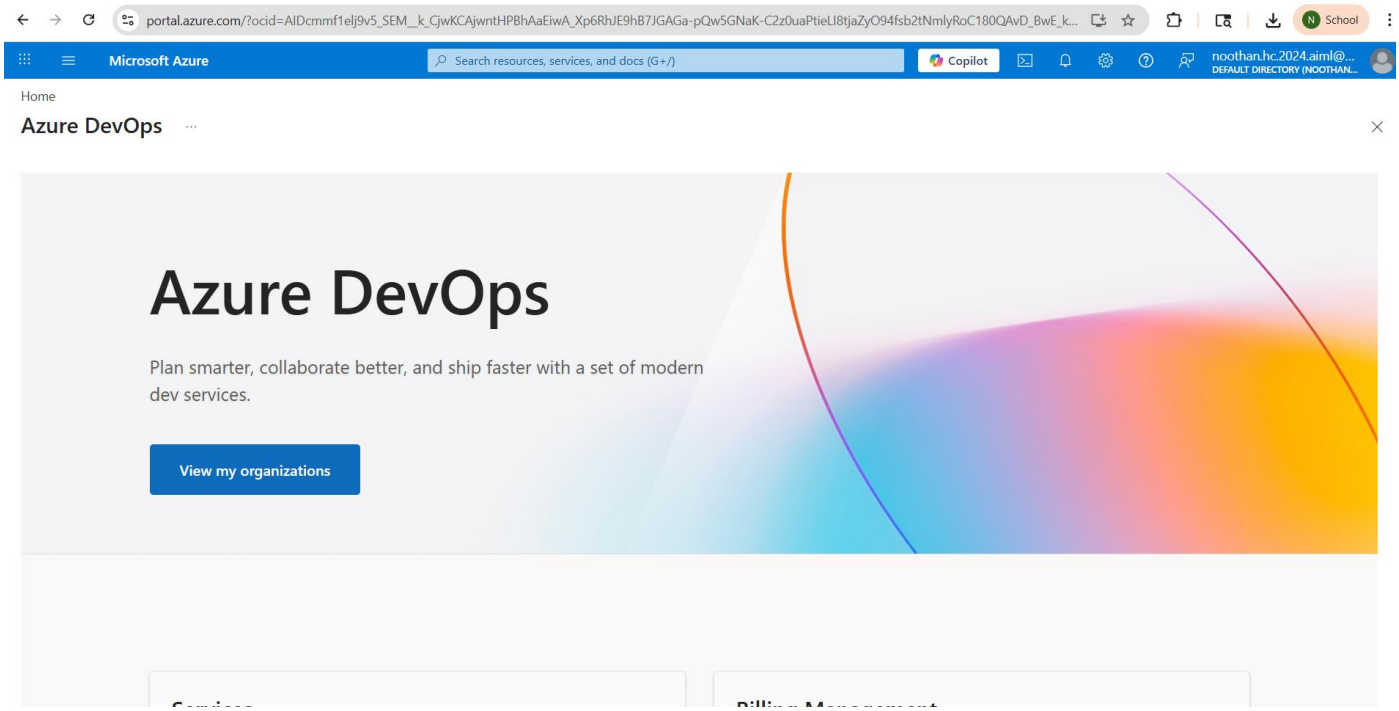
2. Azure home page



3. Open DevOps environment in the Azure platform by typing ***Azure DevOps Organizations*** in the search bar.



4. Click on the ***My Azure DevOps Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



RESULT:

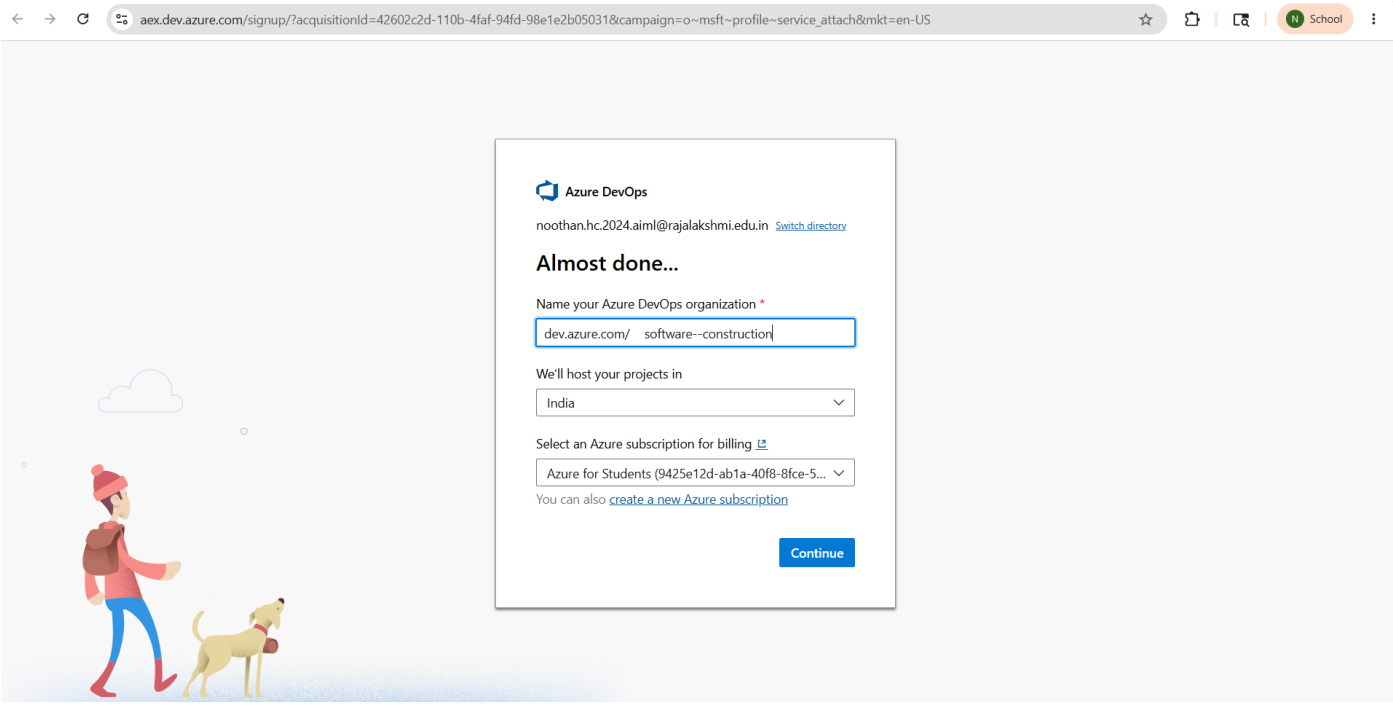
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2 DATE: 30/1/26	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

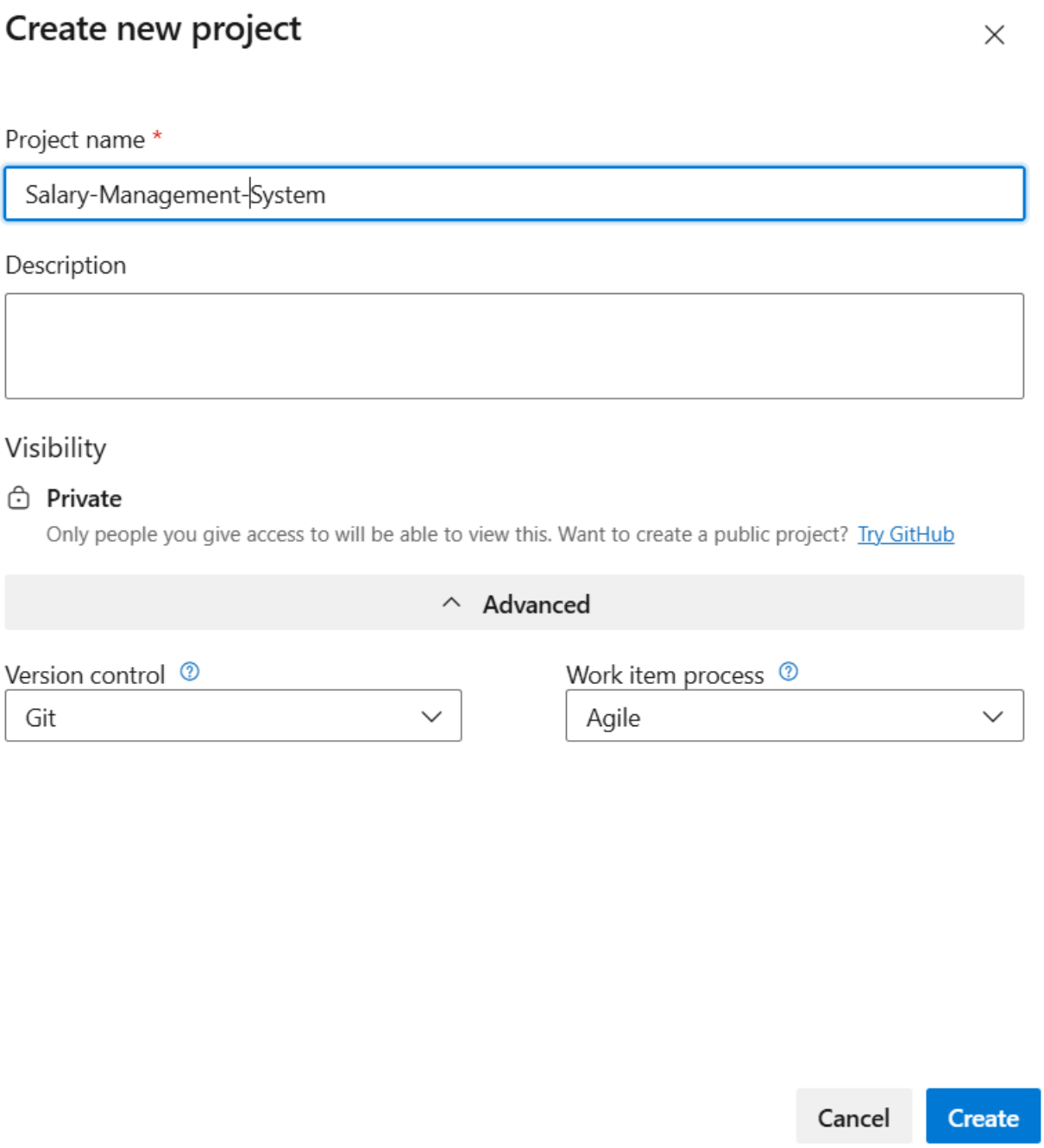
AIM:-

To set up an Azure DevOps project for efficient collaboration and agile work management.

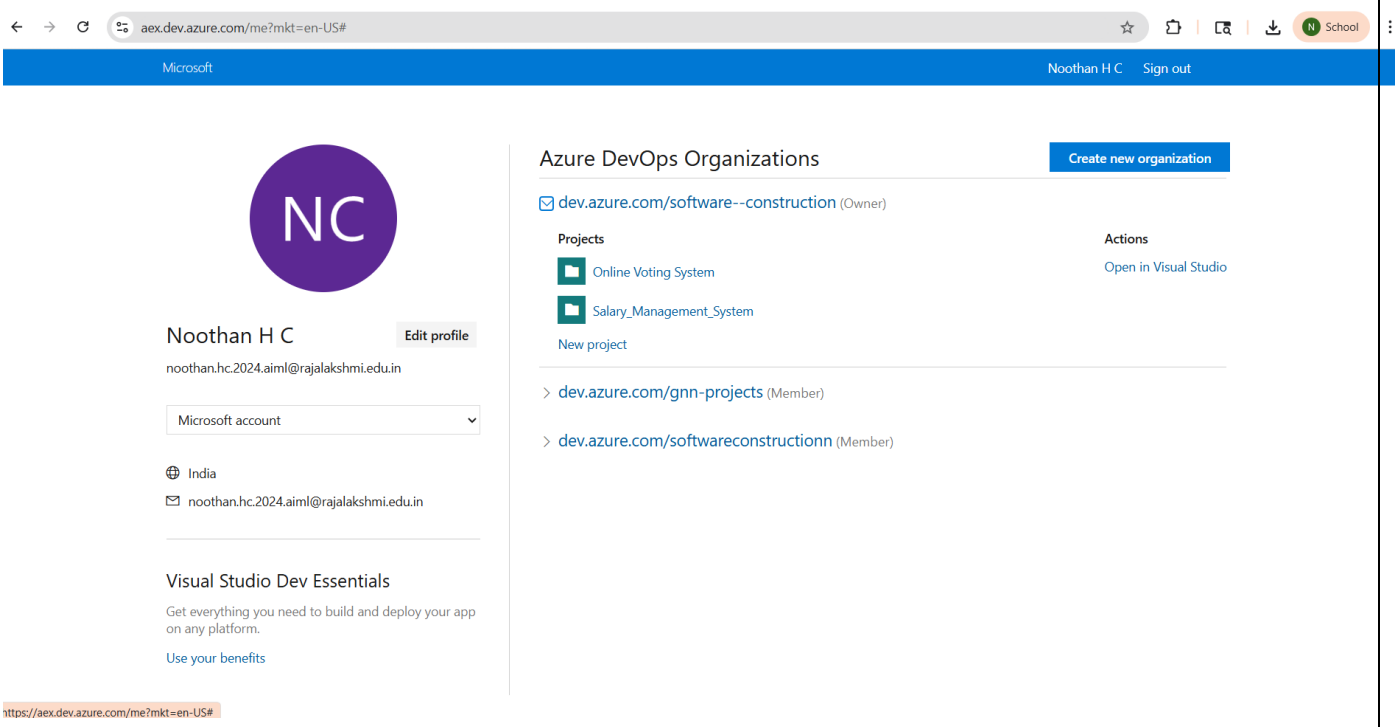
1. Create An Azure Account



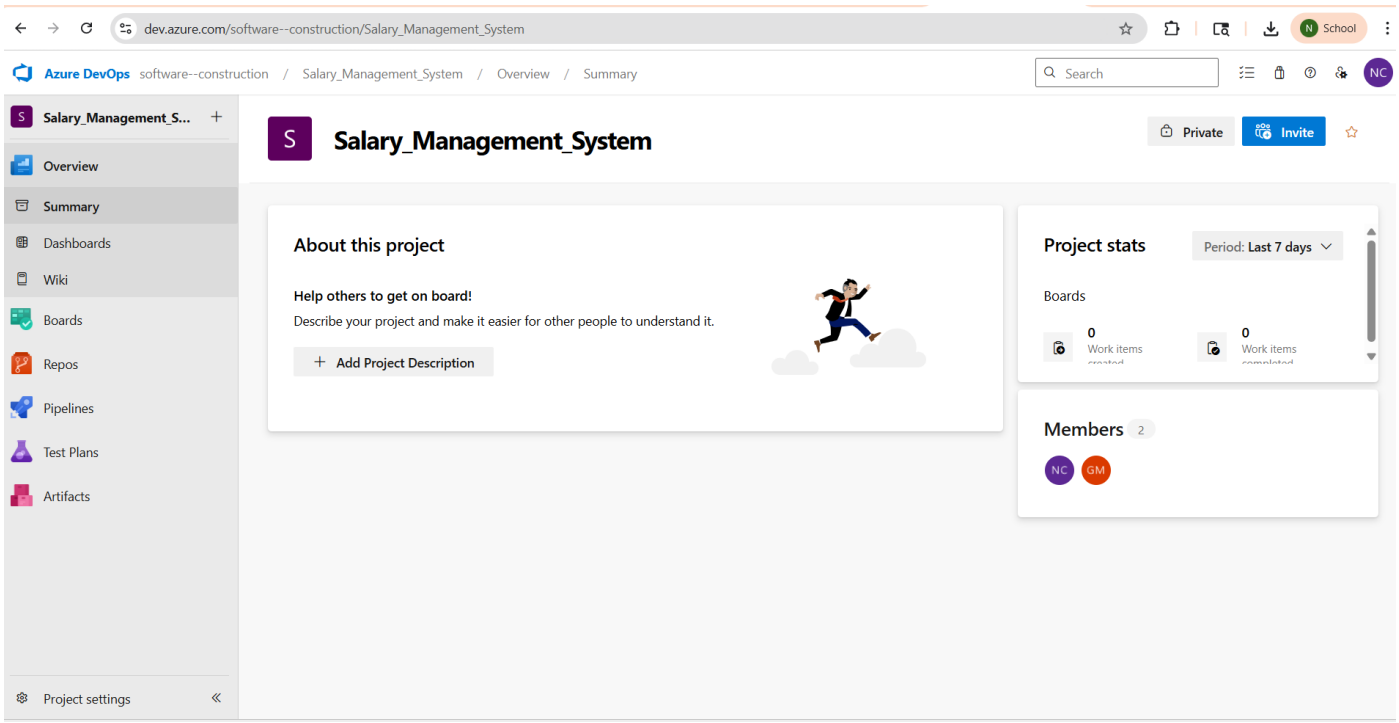
2. Create the First Project in Your Organization
   a) After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
   b) On the organization's **Home page**, click on the **New Project** button.
   c) Enter the project name, description, and visibility options:
      Name: Choose a name for the project (e.g., LMS).
      Description: Optionally, add a description to provide more context about the project.
      Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
   d) Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



4. Project dashboard

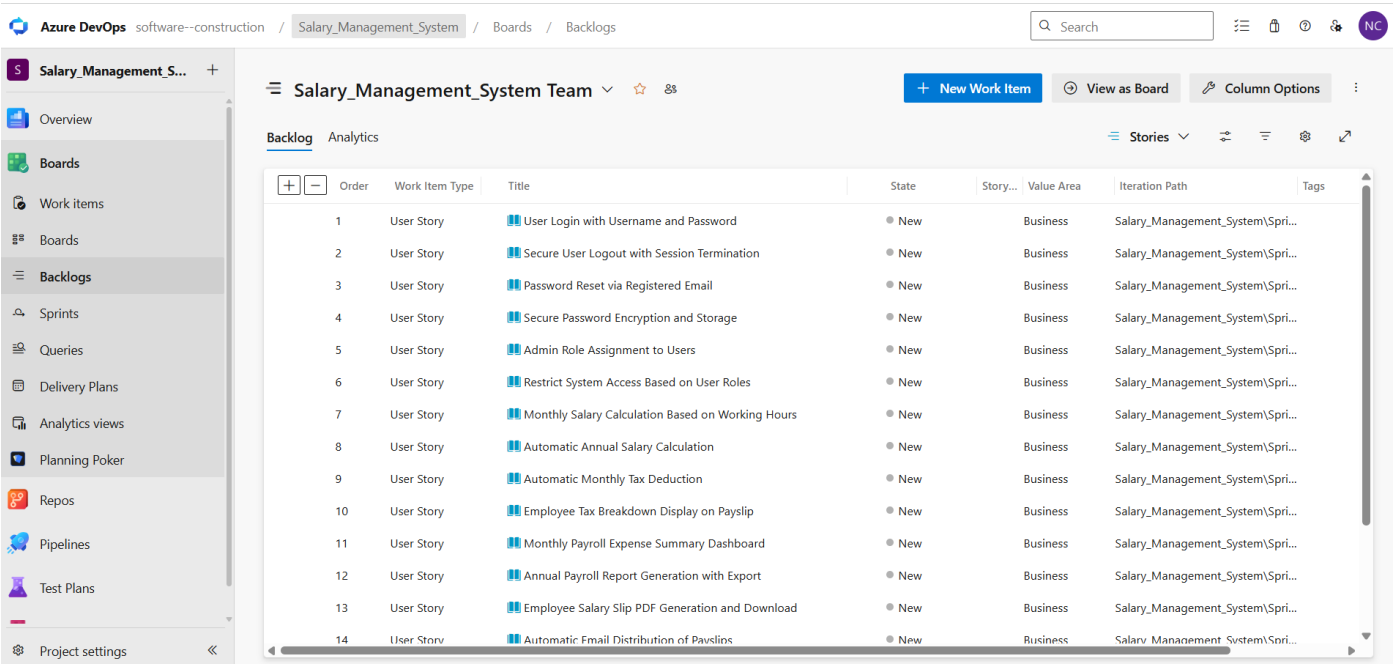


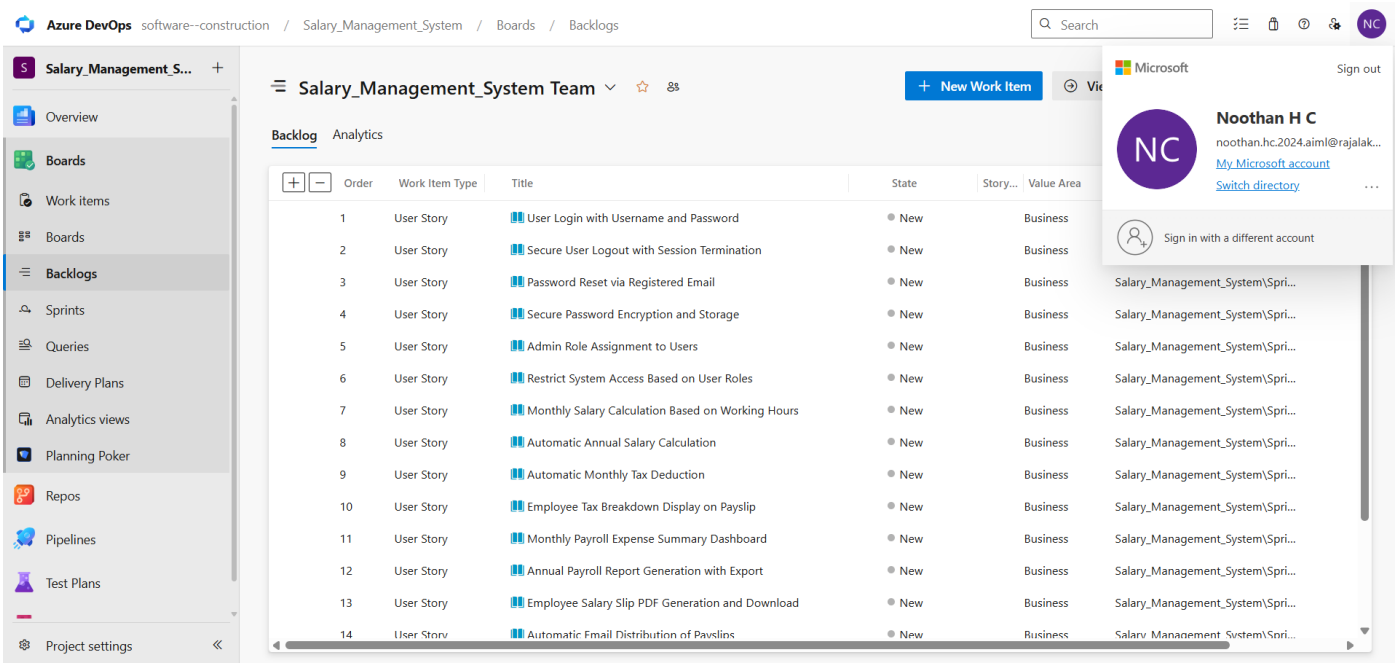
5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards**

page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **Epic**. This will open a form to enter details for the new User Story.





Result

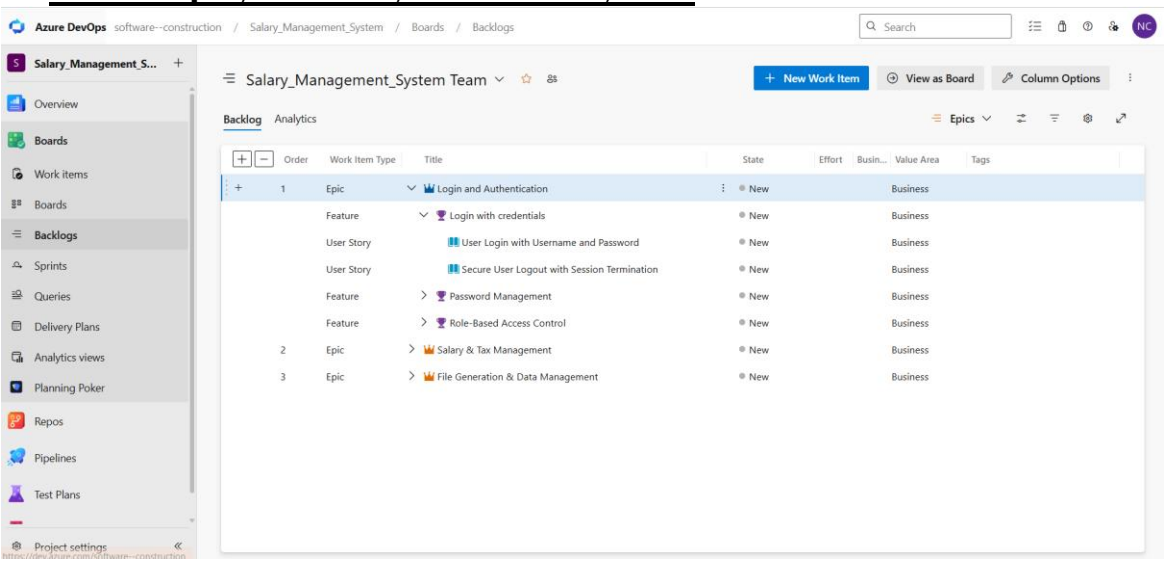
Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3 DATE: 06/2/26	SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

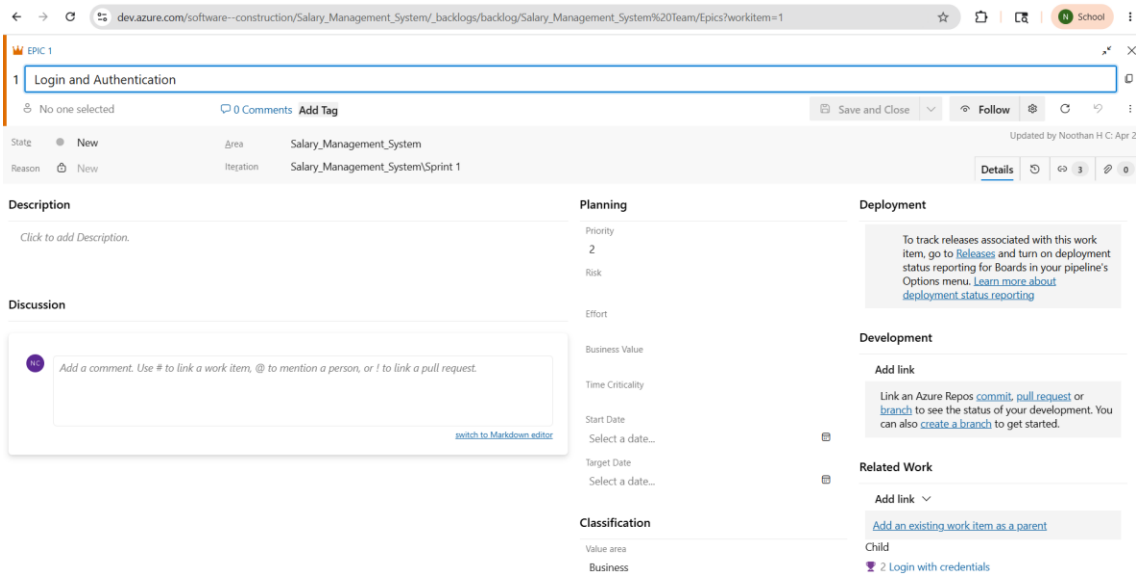
AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

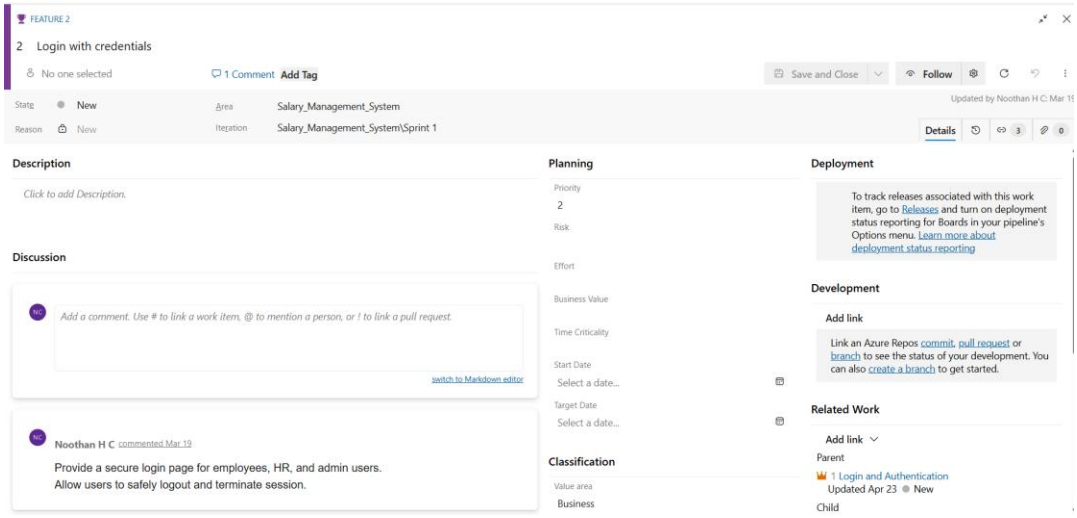
Create Epic, Features, User Stories, Task



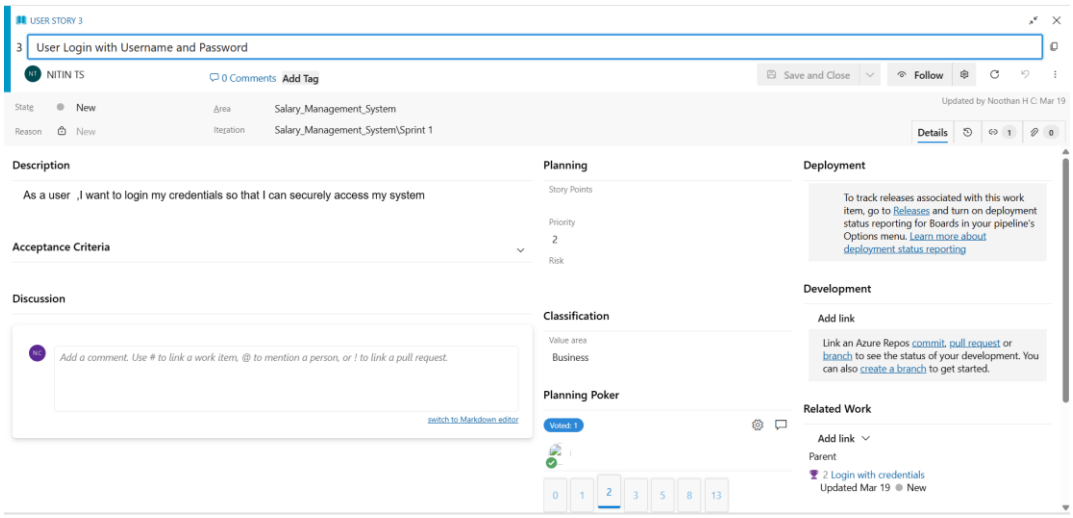
1. Fill in Epics



2. Fill in Features



3. Fill in User Story Details



RESULT:-

Thus, the creation of epics, features, user story and task has been created successfully.

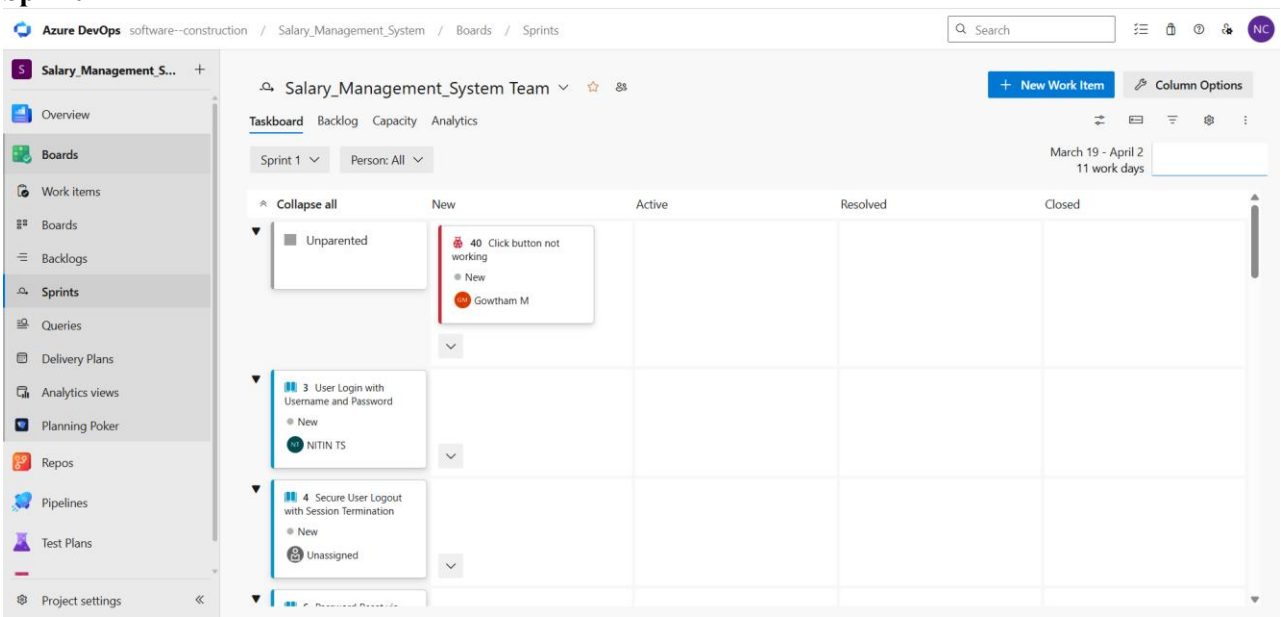
EXP NO: 4 DATE: 13/2/26	SPRINT PLANNING

AIM:-

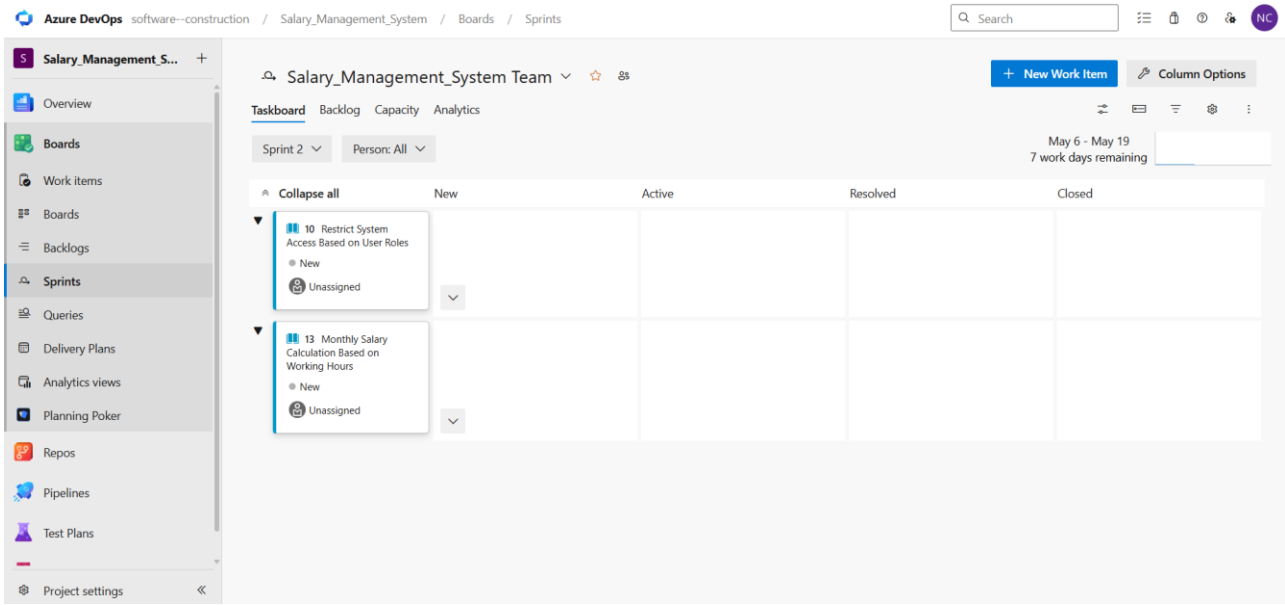
To assign user story to specific sprint for the Student Management System Project.

Sprint Planning:-

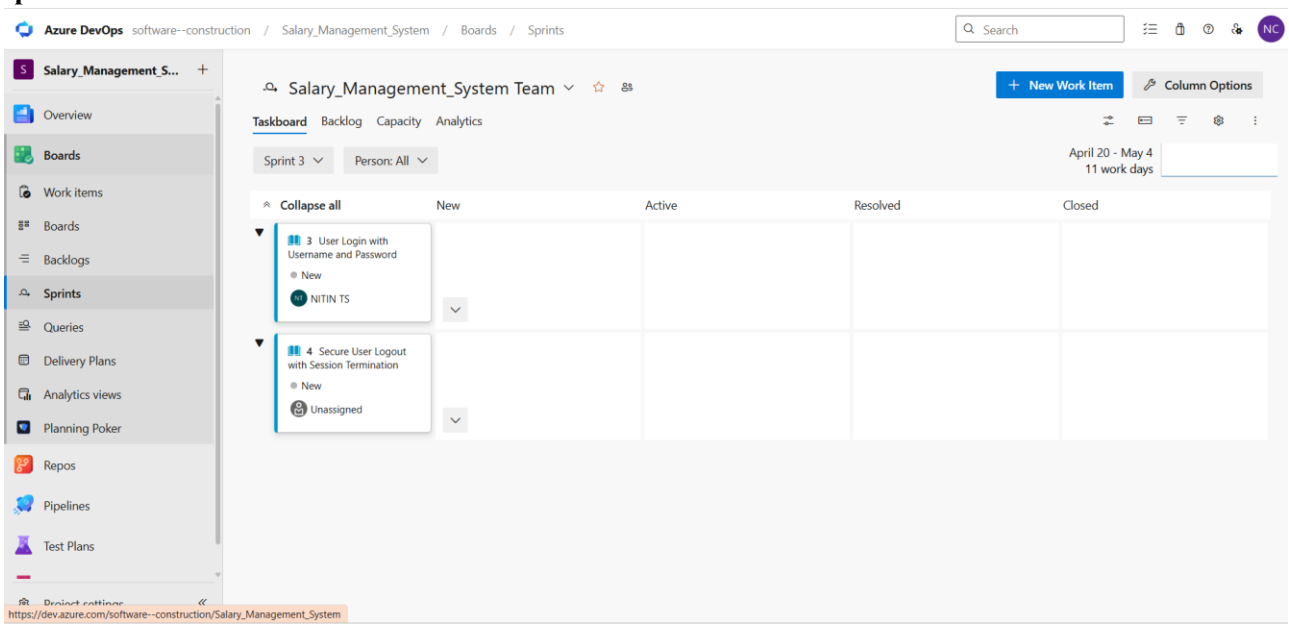
Sprint 1



Sprint 2



Sprint 3



RESULT:-

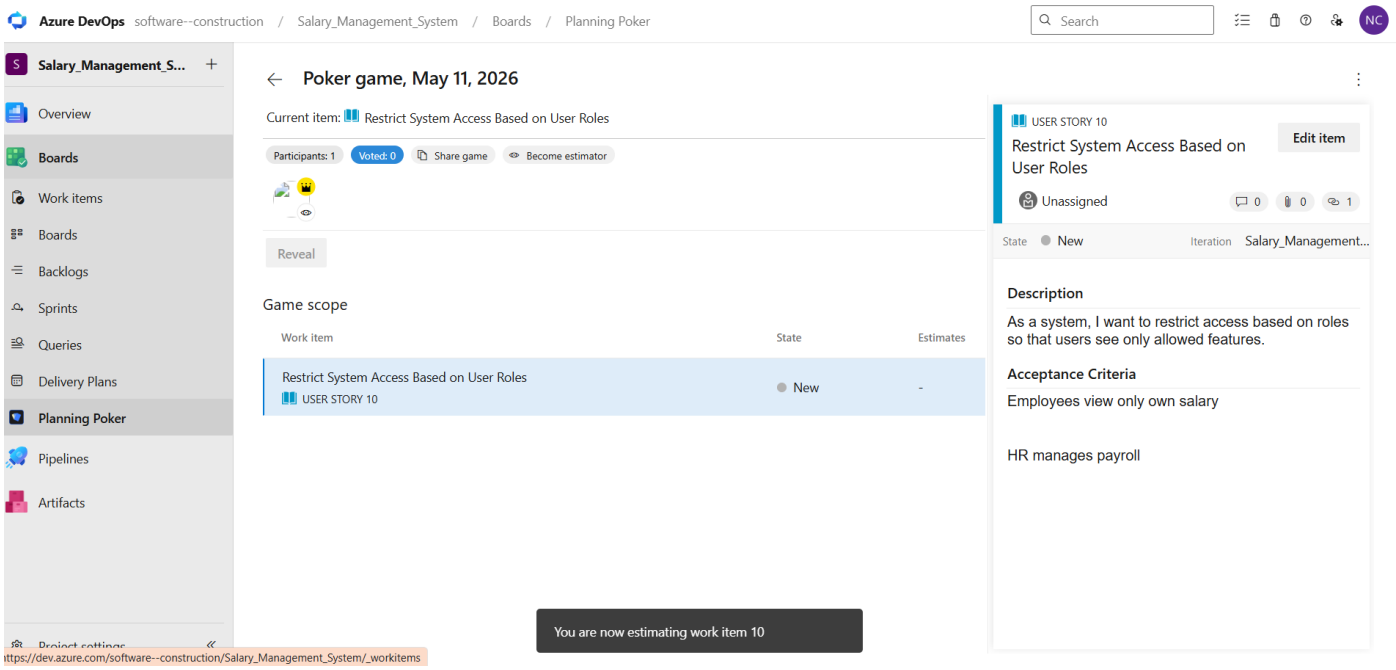
The Sprints are created for the Salary Management System.

EXP NO: 5 DATE: 20/2/26	POKER ESTIMATION

AIM:-

Create Poker Estimation for the user stories -Student management System.

Poker Estimation:-



RESULT:-

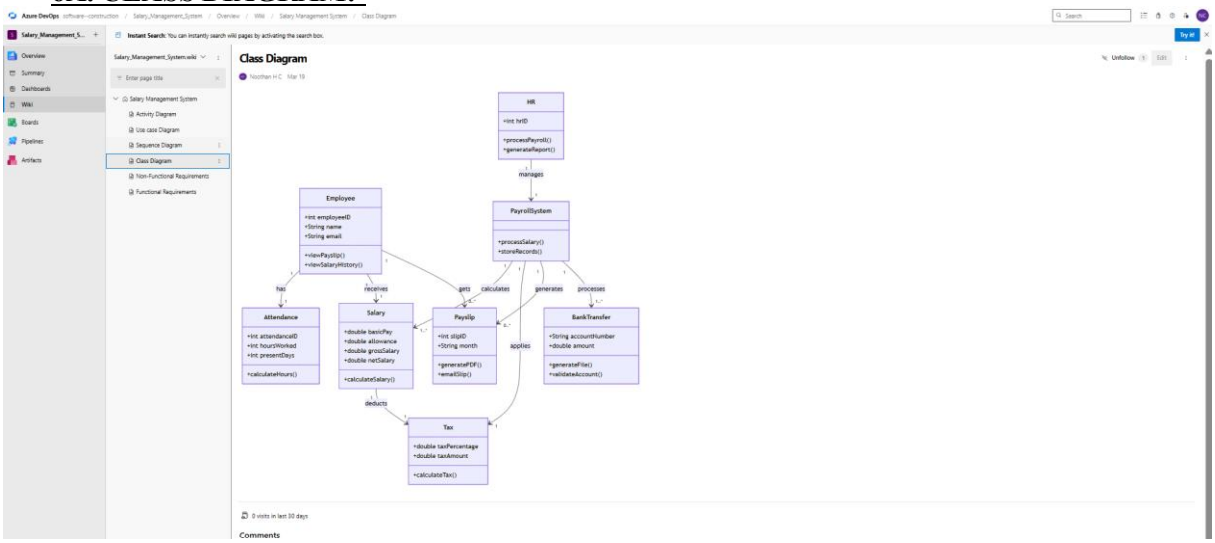
The Estimation/Story Points is created for the project using Poker Estimation.

EXP NO: 6 DATE: 06/3/26	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

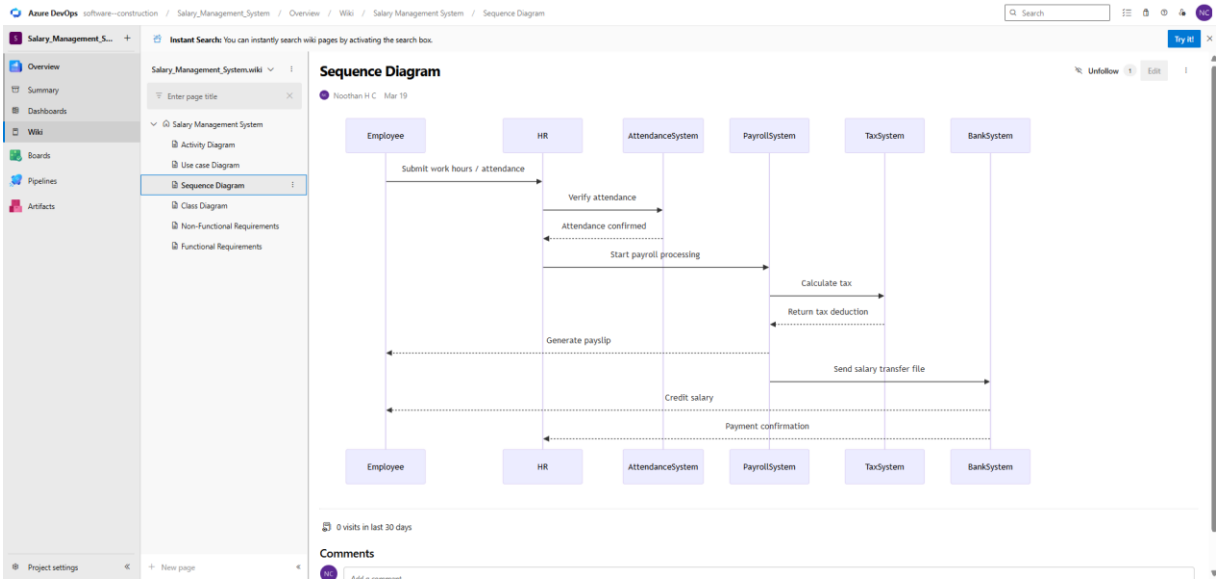
AIM:-

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. CLASS DIAGRAM:-



6B. SEQUENCE DIAGRAM:-



Result:

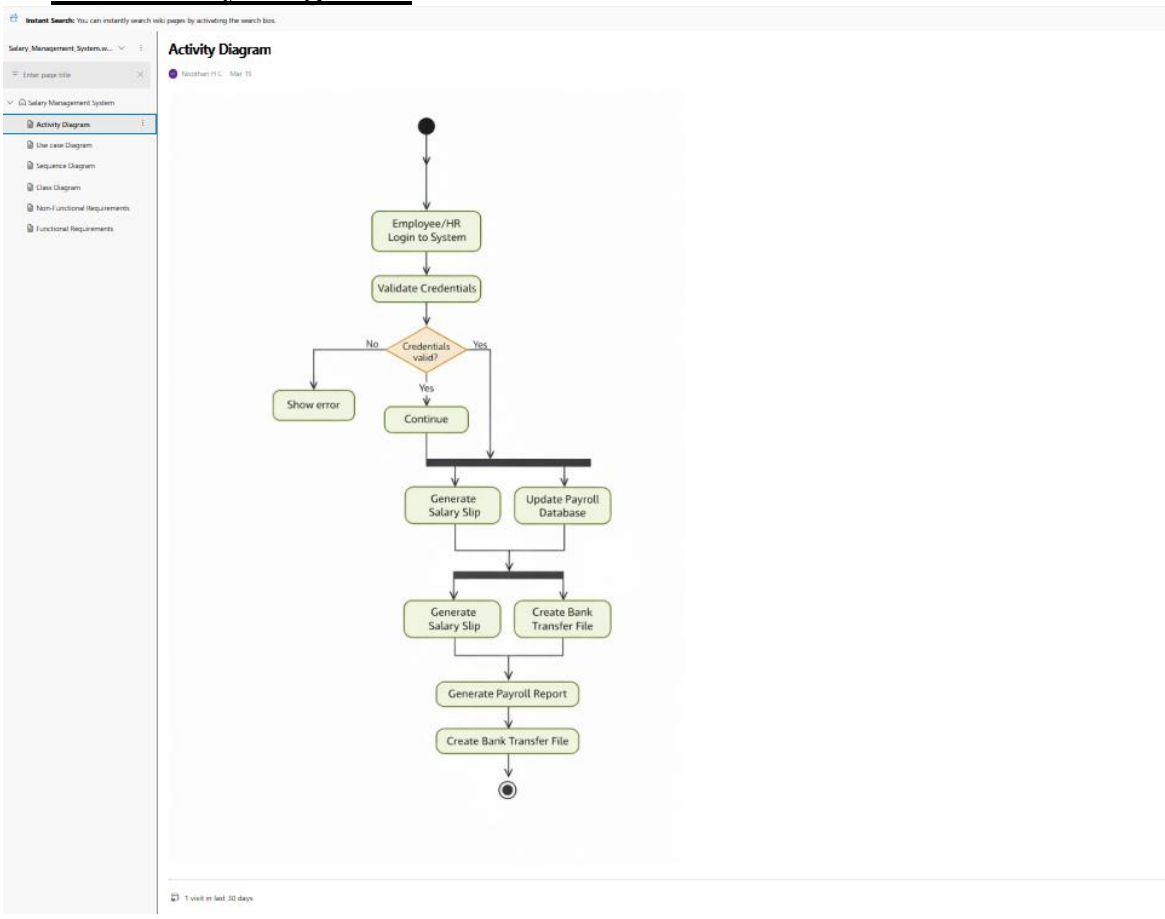
The Class Diagram and Sequence Diagram is designed Successfully for given project.

EXP NO: 7 DATE: 13/03/26	DESIGNING USE CASE DIAGRAMS AND ACTIVITY DIAGRAM FOR PROJECT ARCHITECTURE

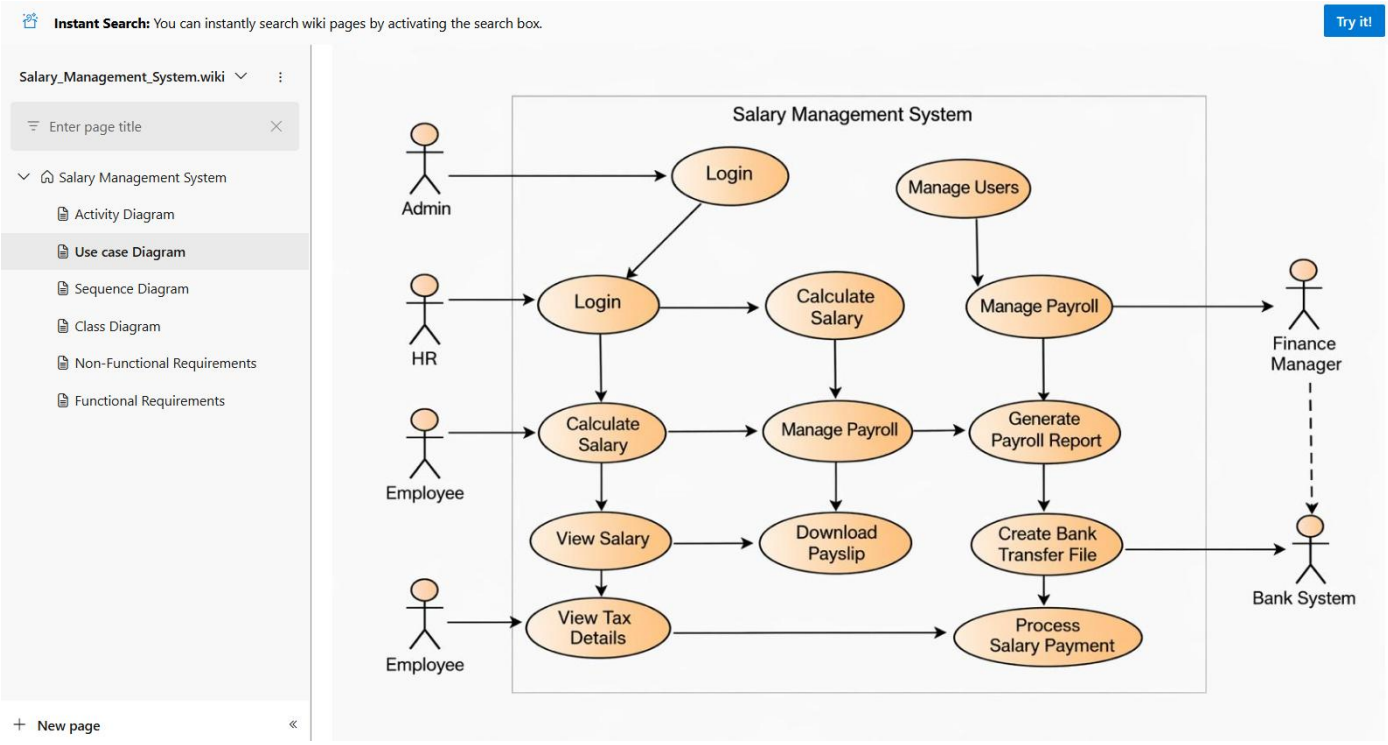
AIM:-

To Design an activity Diagram and use case Diagram for the given Project.

7A. Activity Diagram:-



7B. USE CASE DIAGRAM:-



RESULT:-

Thus activity and use case diagram has been designed successfully for Salary Management System

EXP NO: 8 DATE: 20/3/26	TESTING – TEST PLANS AND TEST CASES

AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case
Test Case Design Procedure

1. Understand Core Features of the Application

- Role-Based Access Control
- Salary & Tax Processing
- Payroll & Report Generation
- File Generation & Data Management

2. Define User Interactions

- Each test case simulates a real user behavior (e.g., logging in, submitting an assignment, viewing results).

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identified
- Facilitates easy mapping to features and tracking in Azure DevOps.

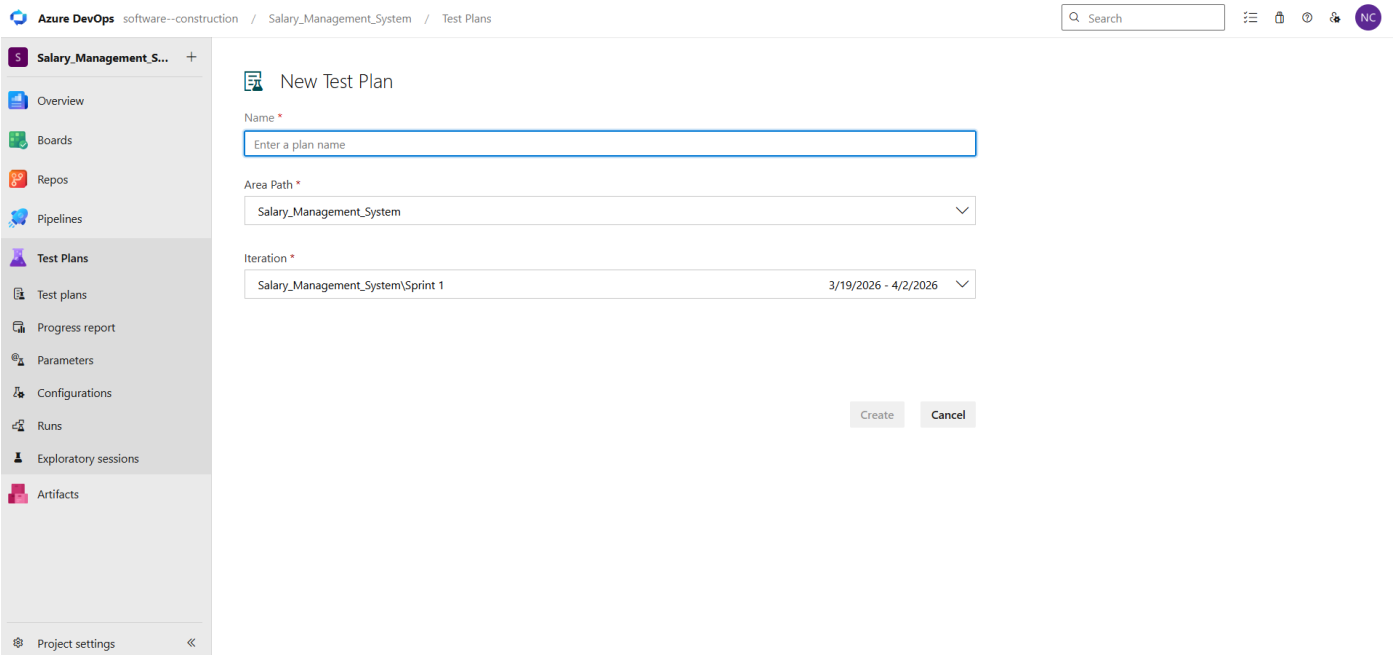
7. Separate Test Suites

- Grouped by functionality such as:
  - Role-Based Access Control
  - Salary & Tax Processing
  - Payroll & Report Generation
  - File Generation & Data Management
- Improves organization and enables focused execution in Azure DevOps.

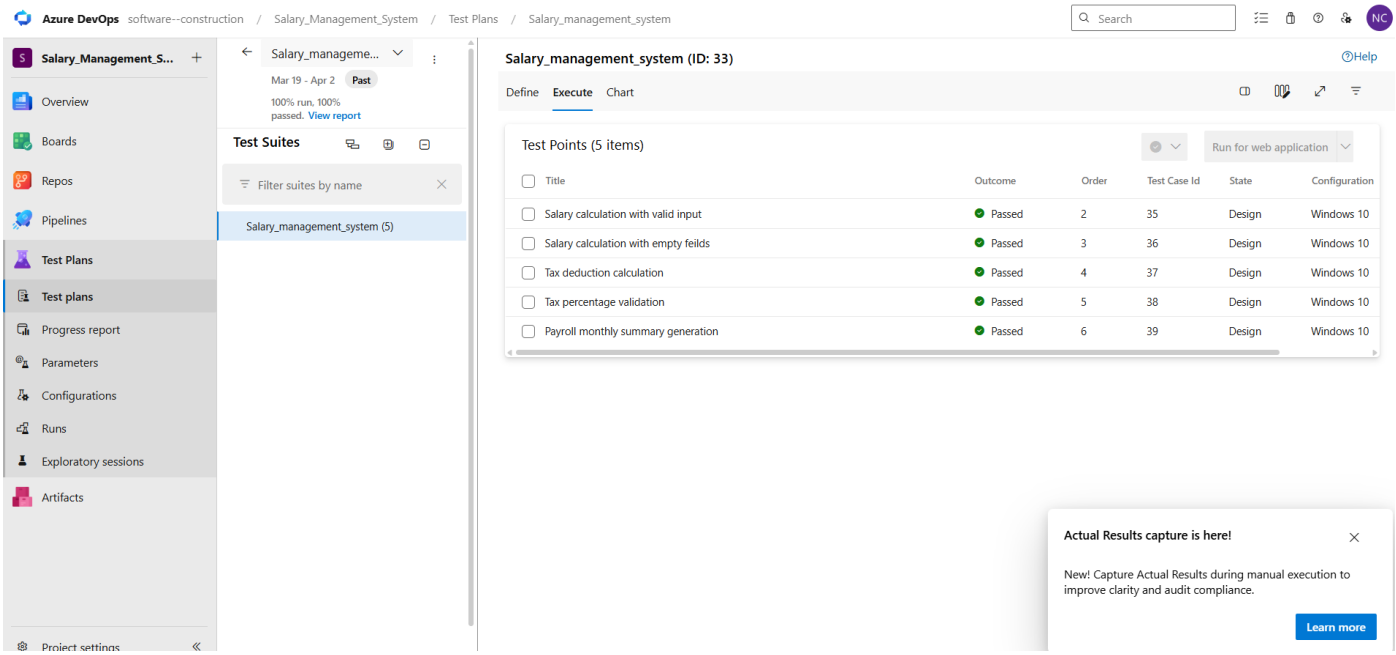
8. Prioritize and Review

- High-priority assigned to critical workflows like login, course access, and grading.
- Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

1. New test plan



2. Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.
Student Management System– Test Plans

USER STORIES:-

• US101	• Login with Credentials	• As a user, I want to login using my credentials so that I can securely access the system.
• US102	• Logout from System	• As a user, I want to logout from the system so that my account remains protected.
• US103	• Reset Password	• As a user, I want to reset my password if I forget it so that I can regain access.
• US104	• Secure Password Storage	• As a system, I want passwords stored securely so that user data is protected.
• US105	• Assign User Roles	• As an admin, I want to assign roles to users so that permissions are controlled.

Test Suites:-

Test suit : TS01 – Login & Authentication (ID: 201)

TC01 – Login with Valid Credentials

- **Type: Happy Path**
- **User Story: US101**

Steps

1. **Open login page**
2. **Enter valid username and password**
3. **Click Login**

Expected Result

- **User successfully logs into system**
- **Dashboard displayed**
- **Session created successfully**

TC02 – Login with Invalid Credentials

- **Type: Error Path**
- **User Story: US101**

Steps

1. **Open login page**
2. **Enter invalid password**
3. **Click Login**

Expected Result

- **Error message displayed: "Invalid username or password"**
- **User remains on login page**

TC03 – Successful Logout

- **Type: Happy Path**
- **User Story: US102**

Steps

1. **Login to system**
2. **Click Logout button**

Expected Result

- **Session destroyed**
- **User redirected to login page**

TC04 – Access Protected Page After Logout

- **Type: Error Path**
- **User Story: US102**

Steps

1. **Login and logout from system**
2. **Use browser back button to access dashboard**

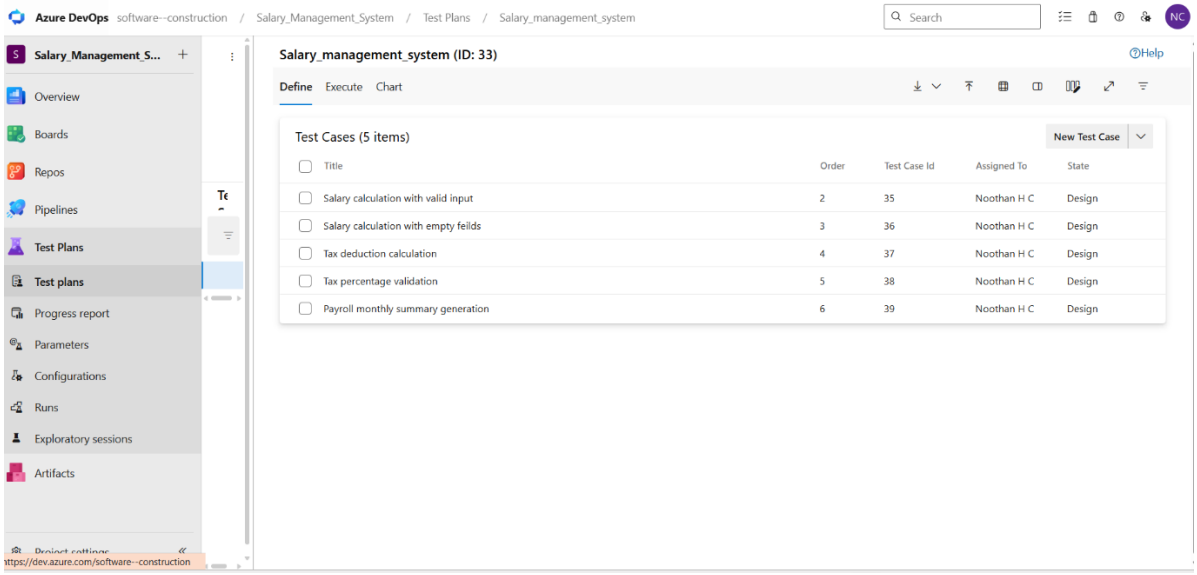
Expected Result

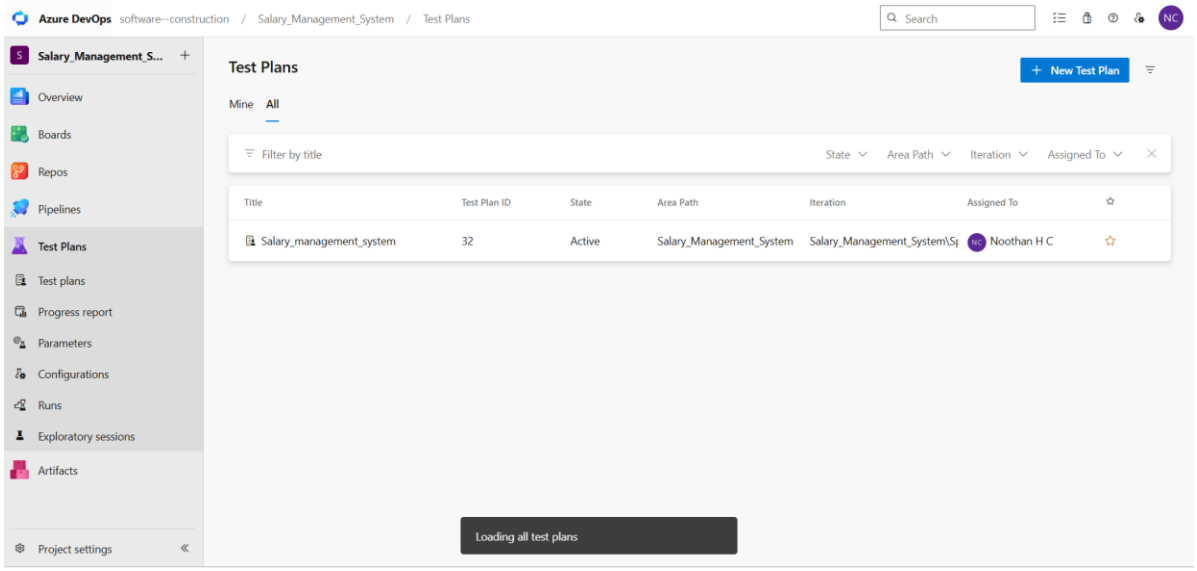
- **User redirected to login page**
- **Protected page inaccessible**

TC04 – Login with Incorrect Password

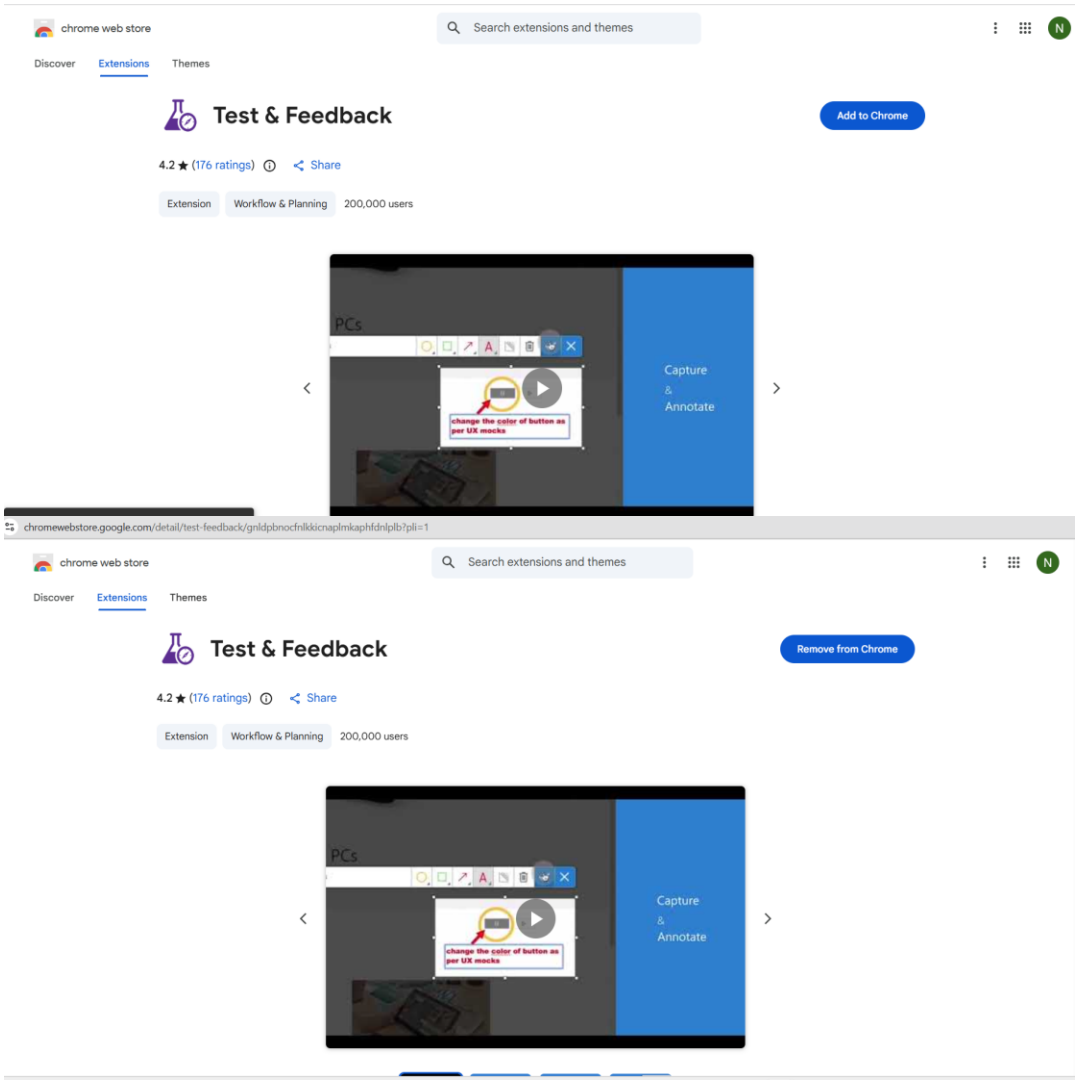
- **Action:**
  - Enter wrong password
  - Click "Login"
- **Expected Result:**
  - Error: "Invalid username or password"
- **Type:** Error Path

1. Test Cases

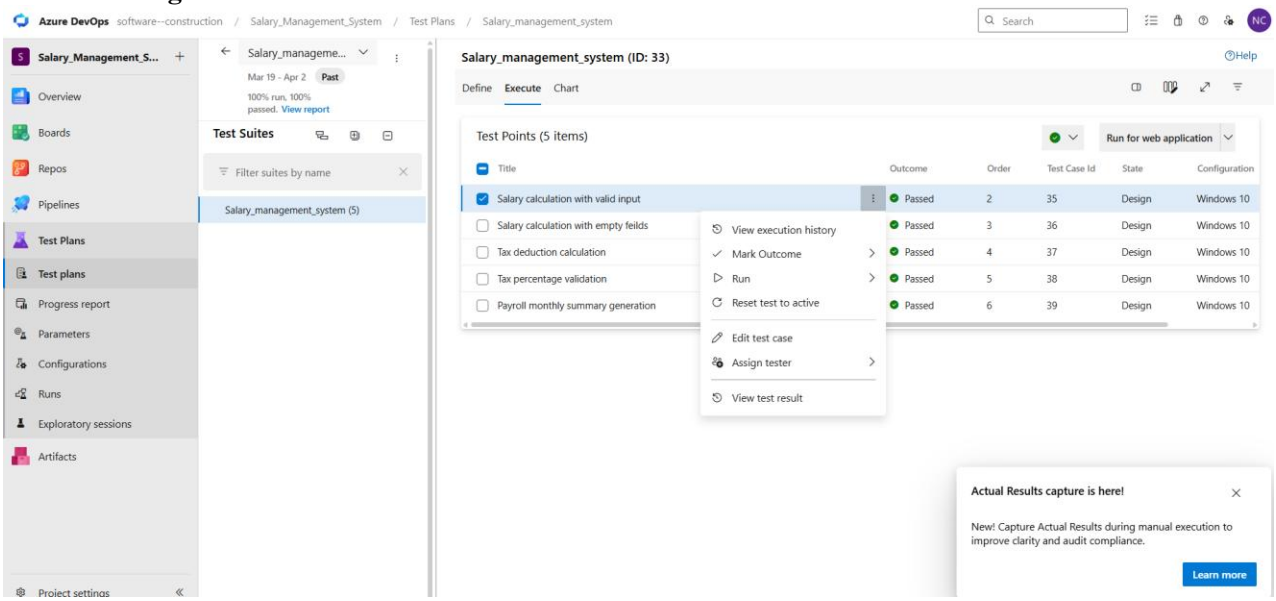

Azure DevOps software--construction / Salary_Management_System / Test Plans / Salary_management_system
Search
Salary_Management_S...
Overview
Boards
Repos
Pipelines
Test Plans
Test plans
Progress report
Parameters
Configurations
Runs
Exploratory sessions
Artifacts
Salary_management_system (ID: 33)
Help
Define Execute Chart
Test Cases (5 items)
New Test Case
Title Order Test Case Id Assigned To State
Salary calculation with valid input 2 35 Noothan H C Design
Salary calculation with empty feilds 3 36 Noothan H C Design
Tax deduction calculation 4 37 Noothan H C Design
Tax percentage validation 5 38 Noothan H C Design
Payroll monthly summary generation 6 39 Noothan H C Design

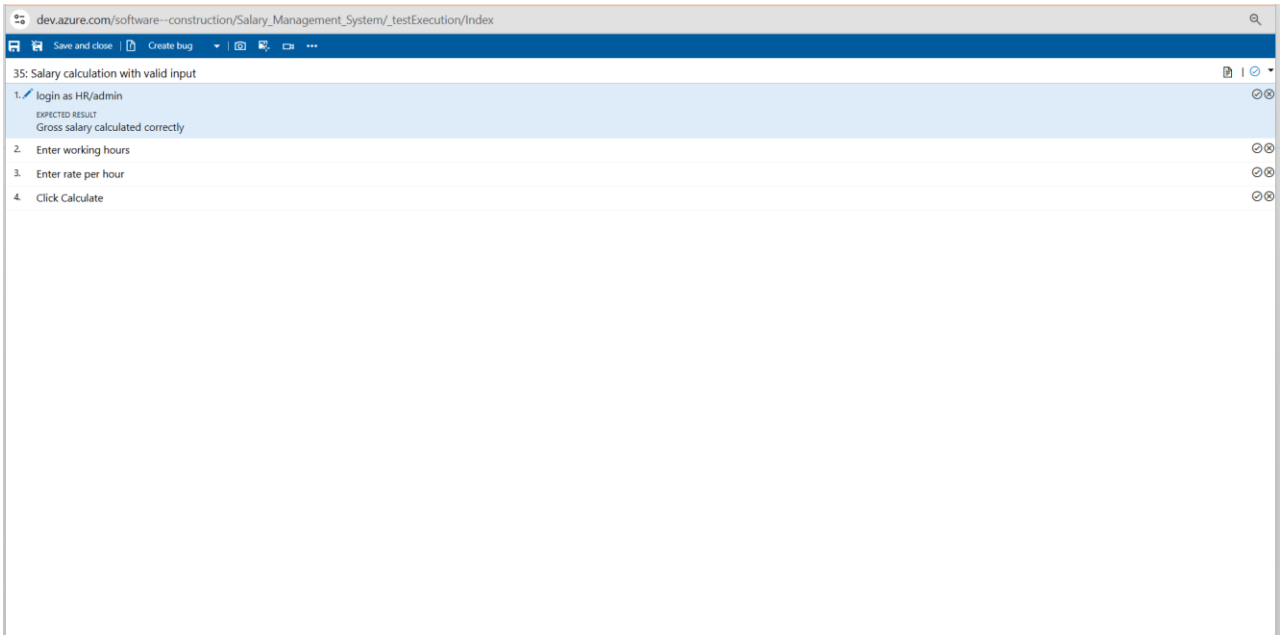

Azure DevOps software--construction / Salary_Management_System / Test Plans
Search
Salary_Management_S...
Overview
Boards
Repos
Pipelines
Test Plans
Test plans
Progress report
Parameters
Configurations
Runs
Exploratory sessions
Artifacts
Project settings
Test Plans
New Test Plan
Mine All
Filter by title
State Area Path Iteration Assigned To
Title Test Plan ID State Area Path Iteration Assigned To
Salary_management_system 32 Active Salary_Management_System Salary_Management_System\S Noothan H C
Loading all test plans

2. Installation of test

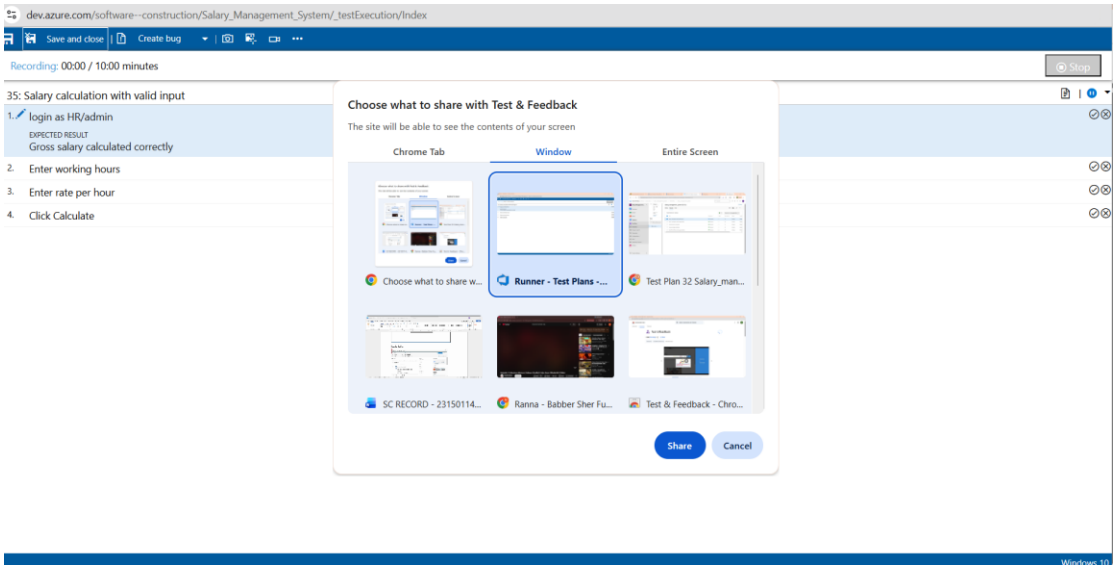


3. Running the test cases

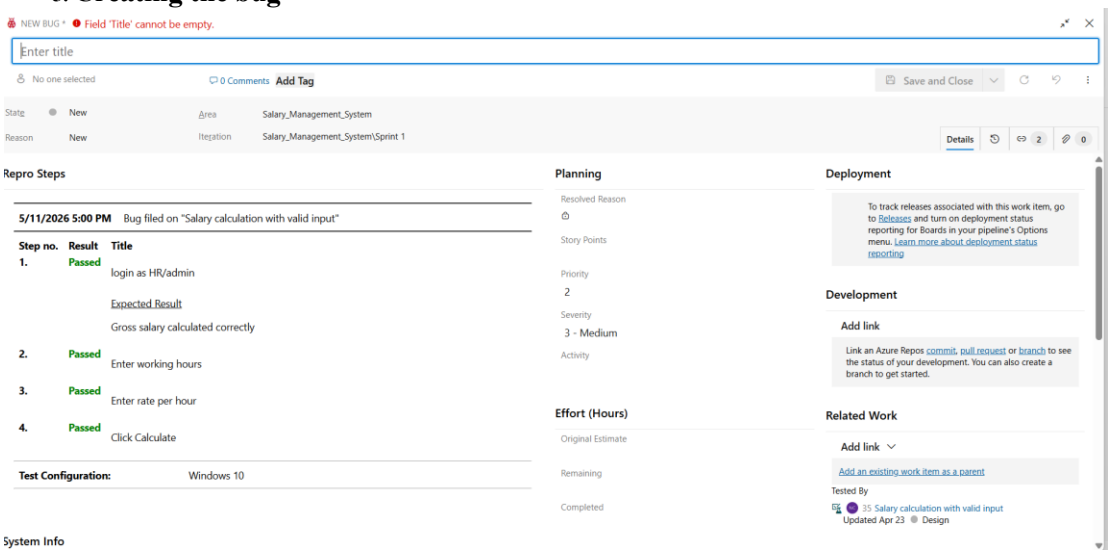




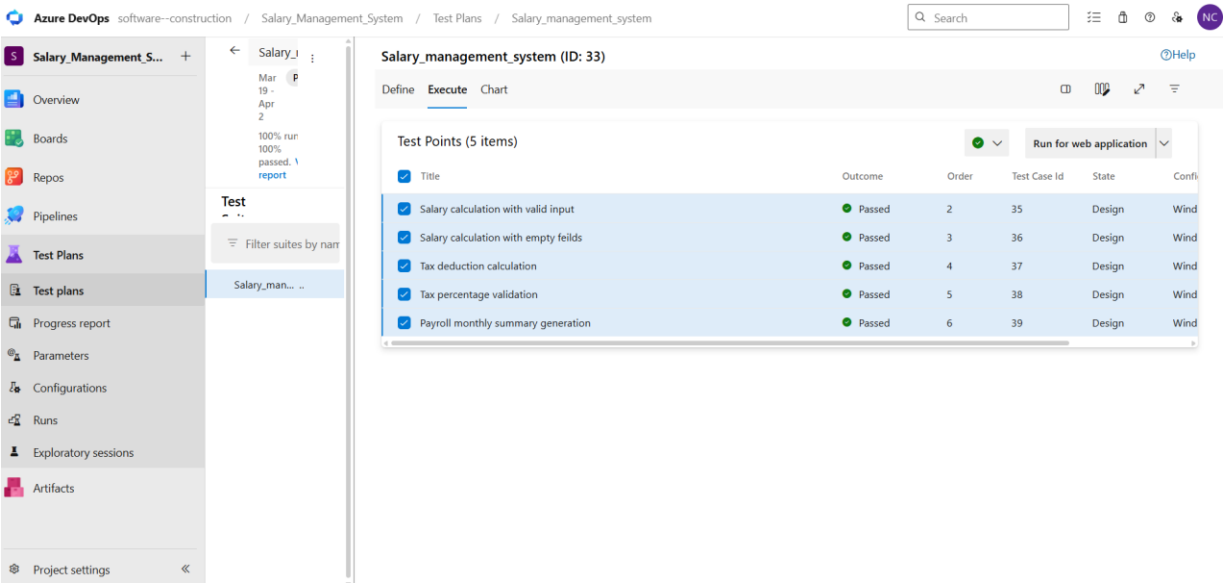
4. Recording the test case



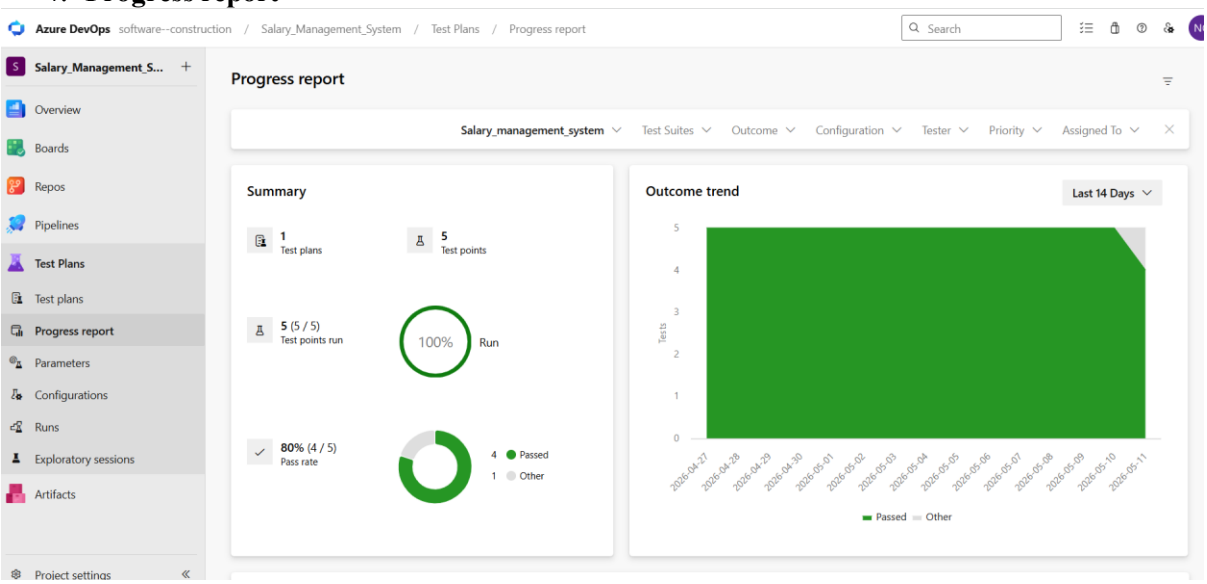
5. Creating the bug



6. Test Results



7. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9 DATE: 10/4/26	CI/CD PIPELINES IN AZURE

AIM:

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the Student Management System, ensuring faster delivery and improved software quality.

PROCEDURE:

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
   Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.

2. Connect a Code Repository (Azure Repos or GitHub)
   Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.

3. Create a New Pipeline
   Go to the Pipelines section on the left panel and click "Create Pipeline".
   Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.

4. Choose the Pipeline Configuration
   You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.
   If using YAML, Azure DevOps will suggest a template or allow you to define your own.

5. Define Build Stage (CI - Continuous Integration) from YAML file

6. Install dependencies (e.g., npm install, dotnet restore)

7. Build the application (dotnet build, npm run build)

8. Run unit tests (dotnet test, npm test)

9. Publish build artifacts to be used in the release stage

10. Save and Run the Pipeline for the First Time
    Save the YAML or build definition and click “Run”.
    Azure will fetch the latest code and execute the defined build and test stages.

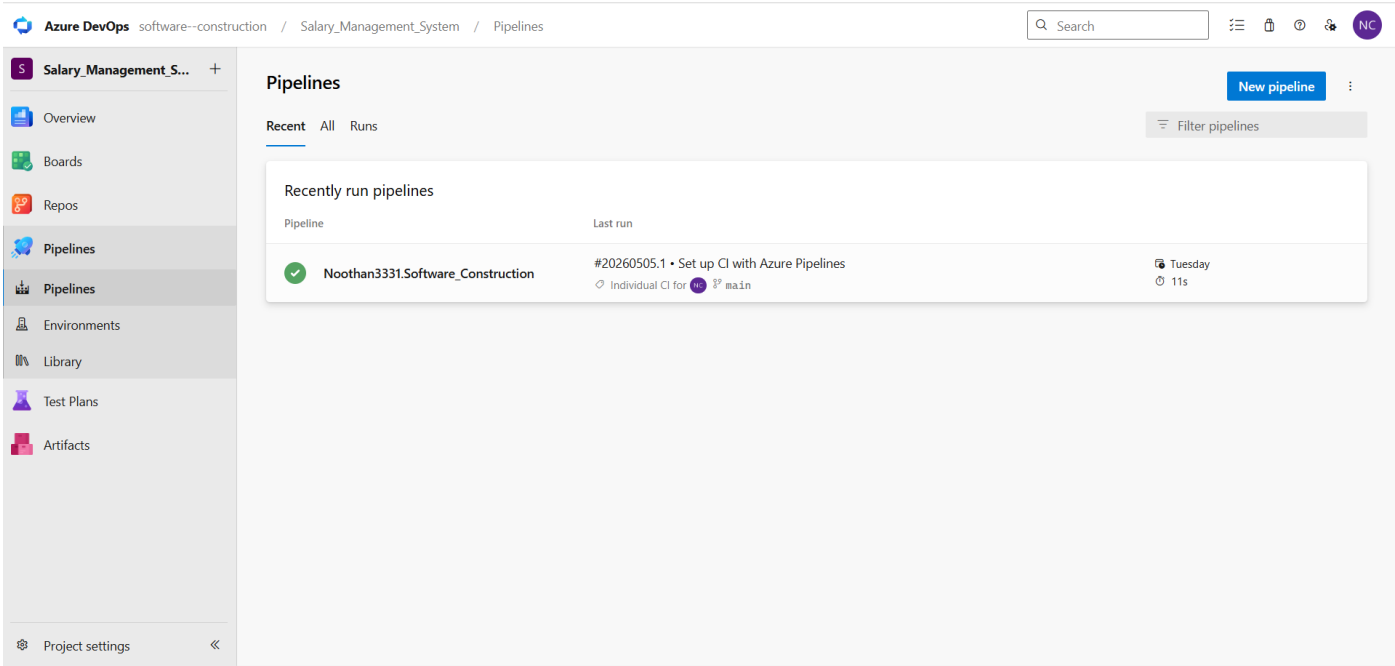
11. Configure Continuous Deployment (CD)
    Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.
    Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals
    Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.
    For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs
    View all pipeline runs under the Runs section.
    Check logs for build/test/deploy stages to debug any errors.
    You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines
    Regularly update your pipeline tasks or YAML configurations as your application grows.
    Ensure pipeline runs are clean and artifacts are stored securely.
    Integrate quality gates and code coverage policies to maintain code quality.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Artifacts
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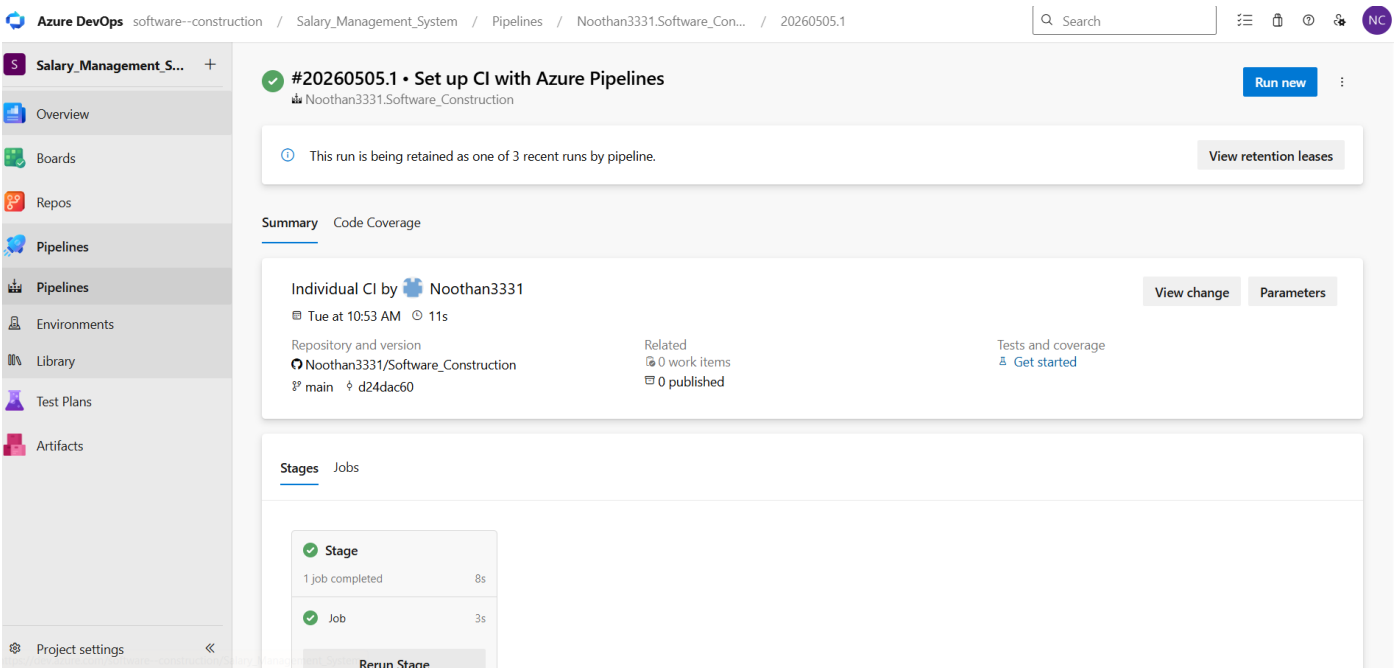
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Noothan3331.Software_Construction
Run new
This run is being retained as one of 3 recent runs by pipeline.
View retention leases
Summary Code Coverage
Individual CI by Noothan3331
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View change
Parameters
Repository and version
Noothan3331/Software_Construction
main d24dac60
Related
0 work items
0 published
Tests and coverage
Get started
Stages Jobs
Stage
1 job completed 8s
Job 3s
Rerun Stage

RESULT:

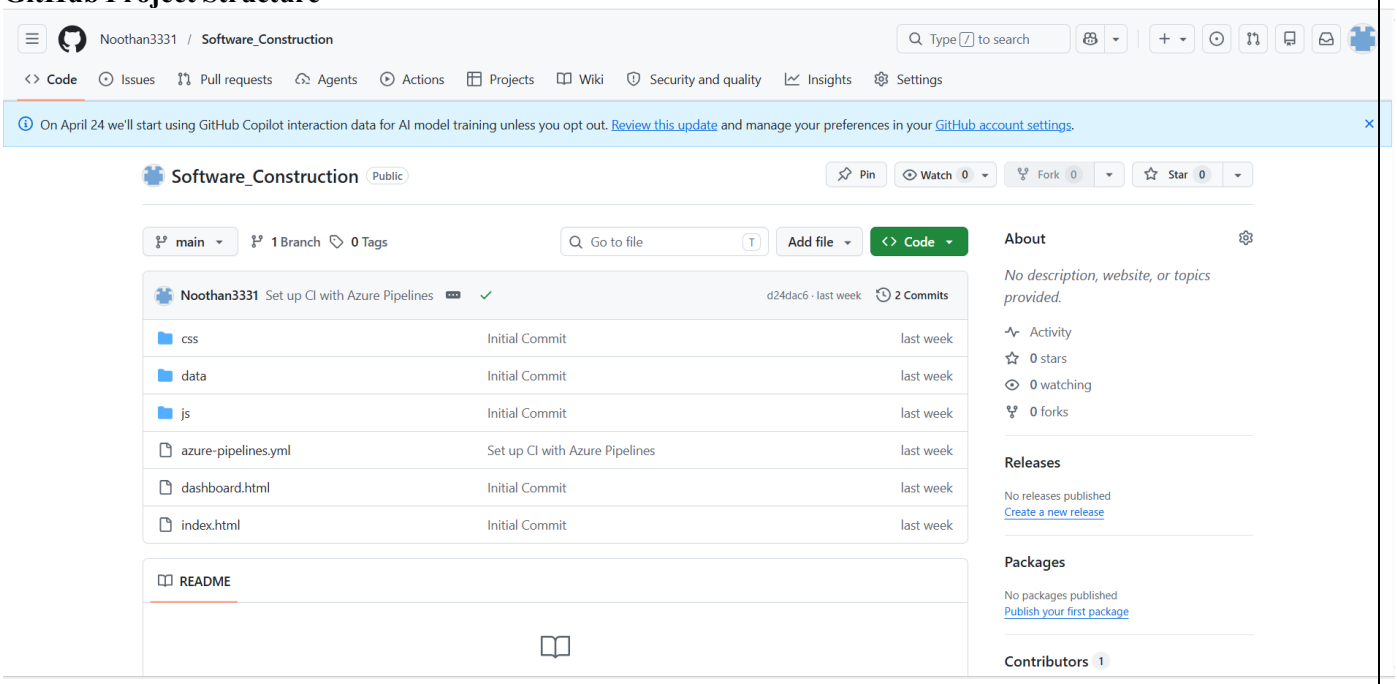
Thus the pipelines for the given project "Student Management System has been executed successfully

EXP NO: 10 DATE: 17/4/26	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Student Management System project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.